Kabir Singh

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Objective

Aspiring physician-scientist in translational medicine: designing peptide therapeutics computationally & modelling biological systems, alongside research in the history of medicine & early modern Sikh history.

Education

B.S. Mathematical Sciences (Mathematical Biology) Sep 2023 – May 2027 (Expected)
Albert Dorman Honors College & Jordan Hu College of Sciences & Liberal Arts, NJIT — Newark, NJ, USA

- Dean's List all semesters.
- Minors: **Biomedical Engineering Technology**, History, Philosophy & Applied Ethics; Honors tracks: Medical Humanities & Global Studies (first Honors College student to pursue two).
- Academic scholarships exceeding $100,000, including Academic Excellence, Honors Merit, Honors Residential, & Highlander Achievement scholarships; tuition & board scholarship support.

Biomedical	**Applications in Healthcare**; Biochemistry; Biomedical Mechatronics; Cell & Molecular Biology
Medicine & society	**Biomedical Ethics**; Health, Justice, & Capitalism; Medical Sociology; **Medicine & Law in Modern History**
Quantitative	Applied Numerical Methods; Differential Equations (ordinary & partial); **Linear Algebra**; **Mathematical Modeling**; Stochastic & Discrete Models in Biology
Independent study	Bioengineering (Dr Vivek Kumar); Maths (Dr James Maclaurin, Dr Horacio Rotstein); History (Dr Stephen Pemberton, Satnam Singh)

Exchange semester (Global E3), Lund University, Sweden Jan 2025 – Jun 2025
Faculty of Engineering (LTH); 44 ECTS — Lund, Sweden

- Fellow, Institute of International Education's Global Engineering Education Exchange (Global E3).

Master's level	Biomathematics; Immunotechnology; Microscopy, Bio-Imaging; **Ultrasound Physics & Technology**
Undergraduate	Patent & Intellectual Property Rights

Research Experience

Computational & Mathematical Biology

Undergraduate Researcher & Computational Biology Team Lead Oct 2023 – Present
KumarLab; PI: Dr Vivek Kumar, University of Houston (previously NJIT) — Houston, TX / Newark, NJ, USA

- Lead undergraduates & graduates in computational development of peptide therapeutics; trained 15+ lab associates & secured over $15,000 in direct project funding.
- Conducted dissections & assisted animal work on angiogenic hydrogels subsequently published; maintained computational research after the lab relocated to Houston.

A novel cyclic peptide that blocks human autoimmune antibodies for the treatment of multiple sclerosis.

- Designed cyclic peptides against multiple sclerosis autoantibodies with eight-residue design scaffold & Rosetta/FlexPepDock workflow; ran molecular-dynamics simulations for my top-ranked peptide CAP8, with GROMACS simulations & alanine scanning; in the published study, CAP8 bound anti-GlialCAM autoantibody with K_d = 0.294 mM, versus 4.82 mM linear & 2.61 mM unrelated cyclic controls & cut epitope detection on Western blot by 69% versus 27% & 5% for controls.
- Partial work through 'independent study' module, published in the *European Journal of Medicinal Chemistry* [1].

A novel cyclic peptide that blocks fertilization protein for a male contraceptive.

- Secured & led a funded non-hormonal male-contraception project: machine-learning-guided cyclic peptide design against *redacted target*, docking & refinement.
- 8 poster presentations, scientific manuscript & patent application in preparation [2].

Lipidated self-assembled peptides that modulate GPCR signaling of CB1 receptor.

- Helped run in-silico validation using NAMD, GROMACS for membrane simulations, aggregation propensity, docking, & persistence studies of peptide.
- Project still underway, simulations in progress, 1 poster presentation.

Undergraduate Researcher, Computational Bionanotechnology Jun 2026 – Present
SoftNanoLab; PI: Prof Stefano Angioletti-Uberti, Imperial College London London, UK

Binder squeeze: mimicking natural binders.

- Apply statistical mechanics, molecular dynamics, protein language models & machine-learning methods to enzyme & binder design & soft-matter/biomaterial systems; specifically developing methodology from previous papers "Mind the Gap" & "Squeezing Enzymes" (preprint) to develop more efficient miniproteins derived from natural binders.
- Project still underway, contributed code to BAGEL (Biomolecular Algorithm for Guidance in Energy Landscape); investigate peptide designs that mimic natural ligands; develop mathematical theory & equations from 'squeeze' & its complement 'grow'.

Undergraduate Researcher, Mathematical Oncology Sep 2026 – Present
PI: Dr Horacio Rotstein, NJIT Newark, NJ, USA

A simplified model of the MAPK/ERK pathway for therapeutic modeling.

- Derive reduced mathematical descriptions & characterize retained dynamical properties for practical implementation.
- 'Independent study' module, project still underway, poster in preparation.

Undergraduate Researcher, Mathematical Modeling Oct 2025 – May 2026
PI: Dr James Maclaurin, NJIT Newark, NJ, USA

Modeling amyloid as an epidemic.

- Applied Hawkes self-exciting point processes in reframing Aβ42 secondary nucleation dynamics & feedback in amyloid formation.
- 'Independent study' module, project complete, internal paper for course written.

Intellectual & Cultural History

Emerging Scholars Research Fellow, Early Modern Sikh History Sep 2026 – Dec 2026
Harvard Sikh Center; mentor: Satnam Singh Remote

Dissemination and the making of Khalsa authority in a contested panth.

- How the early Khalsa came to dominate the wider Sikh world against rival claimants, including the Bandai Khalsa, Minas, Ram Raiyas & Hindaliyas.
- Argue that the answer lies less in politics than in transmission: the Khalsa's consolidation of 'authentic Sikh teachings' through *katha* preceded & enabled its control of the panth.
- Sources: Gurmukhi & Persian primary materials, including works attributed to Bhai Mani Singh, *hukamname*, *rahit-nama* & sampardaic literature, alongside Persian & European accounts of the period.
- Twelve-week fellowship term producing a paper & a presentation: first, a characterization of the intellectual landscape & its conflicts; second, a prosopography linking intellectual influences to the period's political figures.

Undergraduate Researcher, History of Medicine & Bioethics Sep 2026 – Present
PI: Dr Stephen Pemberton, NJIT Newark, NJ, USA

Physician, heal thyself: hidden death in American medicine.

- Trace the legal history of end-of-life decision-making from *In re Quinlan* (1976) to *Cruzan v. Director, Missouri Department of Health* (1990).
- Compare how three physician-authors, Sherwin Nuland, Atul Gawande & Haider Warraich, account for dying in modern American medicine.
- 'Independent study' module, project complete, academic manuscript in preparation.

Undergraduate Researcher, Digital History & Urban Studies Jan 2024 – May 2024,
PI: Dr Louis Hamilton, NJIT; field research in Rome & Florence Jan 2026 – May 2026
Newark, NJ, USA / Italy

- Two solo-authored geospatial narratives (StoryMapJS) mapping sites visited on foot against historical & archival sources.

Power vs. the People: the Via Giulia, Rome.

- Mapped 18 sites, layering Counter-Reformation Marian shrines against Julius II's 1508 papal axis along the Via Giulia, Rome, drawing on the Early Modern Rome Reconstructed database & Rufini's 1853 shrine census; argued the street shows superimposition & negotiation rather than a papal-versus-popular divide.
- Course project (HIST 328), complete; published online as a StoryMap.

Processions and Power: Florence.

- Mapped 22 Florentine sites to examine how urban expansion transformed liturgical processions into contests of ecclesiastical & household prestige, read against Dante's critique of the *gente nuova*.
- Course project (HIST 329), complete; published online as a StoryMap.

Emergency Care & Health Services

Undergraduate Researcher, Collegiate Emergency Medical Services Sep 2025 – Present
PI: Dr Chang Yaramothu, NJIT Newark, NJ, USA

A comprehensive study of operational models in university based EMS organisations.

- Co-developed IRB protocol, questionnaire & recruitment materials for a national study of collegiate EMS organizations; examined resources, organizational structures, operating models & capabilities across US collegiate EMS squads.
- Data collection & analysis underway.

Publications & Manuscripts

[1] C. Heffernan, O. West, **K. Singh**, J. Dodd-o, M. James, A. Baradi, K. Rybak, A. Lateef, S. Elkabes, V. Kumar. "A novel cyclic peptide that blocks human autoimmune antibodies for the treatment of multiple sclerosis." *Eur. J. Med. Chem.* (2026). doi:10.1016/j.ejmech.2026.119173
Contribution: ran the molecular dynamics simulations; designed the published peptide

[2] S. Saxena, **K. Singh**, M. Young, A. Pallera, C. Brewer, K. Carpenter. "Impact of a youth-focused microgrant on undergraduate and graduate research." In preparation

[3] **K. Singh**, C. Heffernan, O. West, J. Dodd-o, V. Kumar. "Machine learning for rational design of a cyclic peptide-based male contraceptive." In preparation; patent application in preparation

[4] J. Lau, D. Yun, **K. Singh**, A. Patananan, N. Jadavji. "Academic publishing as a launchpad: how publishing research in an undergraduate journal shapes future careers." In preparation

Presentations

M. Kotta, G. Victorio, G. Karmacharya, **K. Singh**, N. Giovannetti, B. Shadpoor, J. Dodd-O, V. Kumar. "Putative computational design of novel inhibitors to modulate de-novo targets for CB1 receptor."
University of Houston, Undergraduate Research Day (*Poster & podium*) 2026

K. Singh, C. Heffernan, J. Dodd-O, V. Kumar. "Development of a novel male contraceptive using cyclic peptides to inhibit fertilization."

AAP/ASCI/APSA Joint Meeting (*Poster*) 2026
NJIT AI Exploration Day (*Poster*) 2026
NJIT Honors Interdisciplinary Research Forum, first place (*Poster*) Fall 2025
National Contraception Meeting (*Poster*) 2025
NJIT Honors Interdisciplinary Research Forum (*Poster*) Spring 2025
NJIT Dana Knox Research Showcase (*Poster*) 2025
NJIT Undergraduate Research & Innovation (*Abstract & podium*) 2025
Biomedical Engineering Society (BMES), co-presenter (*Abstract & poster*) 2024
NJIT Undergraduate Research & Innovation (*Abstract & podium*) 2024

K. Singh, J. Lau, H. Gerber, A. Alibilli, L. Goralsky. "Mitigating knowledge drain: a case study in optimizing workflow and training in a high-turnover, international peer-reviewed undergraduate science academic journal."

National Conference on Undergraduate Research (NCUR) (*Poster*) 2026

Lectures & Features

Invited speaker (forthcoming), Science National Honor Society – Piscataway High School 2026

Invited student roundtable speaker 2026
- Margo Chaly, New Jersey Secretary of Education; NJIT

Invited speaker, NJIT University Advancement Dinner 2026
- Interdisciplinary medicine & healthcare research for NJIT leadership, including interim president & interim provost

Speaker, 'Shaping the Future of Healthcare & Innovation' dinner Oct 2026
- Peptide therapeutics, published multiple sclerosis research, & emergency care; alongside Dr Genoa R. Warner & Dr Bryan Pfister, NJIT

Lecture series, Ethics in Science and Research – Science Mentorship Institute 2026

Session moderator, Student Oral Presentations (Communications) – NCUR 2026

Featured presenter, 'Navigating peer review and the student journal landscape' 2026
- 'Shaping Asia's Future: Development, Innovation, and Society in Transition', Nazarbayev University / Asian Universities Alliance

Featured panelist, 'It's Different for Girls' 2026
- Documentary screening & panel; NJIT ADHC & CSLA co-sponsored colloquium

NJIT Makes: Undergraduate Research and Innovation – NJIT promotional material 2026

Invited speaker, NJIT Board of Trustees and President's Town Hall 2025
- Presented with NJIT First Aid Squad leadership at the request of President Teik Lim

Honors in Action 2025 2025
- Albert Dorman Honors College publication: research, awards & study abroad

Article feature: '150+ Honors Scholars Showcase Findings at Interdisciplinary Research Forum' – Nayib Morán, NJIT News 2025

Featured speaker, ADHC Advising Day – NJIT; appearances in both 2024 & 2025 2024, 2025

Honors in Action 2024 2024
- Albert Dorman Honors College publication: research & study abroad

Article feature: 'Undergraduate Student Researchers on Campus' 2024
- Gurnoor Kaur, *The Vector*, NJIT student newspaper

Lecture, 'Introduction to PyMOL and Linear Peptide Design' – NJIT, Genetically Engineered Machines 2024

Research Funding & Honours

Grants & awards

Undergraduate Research Opportunities Programme (UROP) bursary; ~£100 Jun – Jul 2026
- Imperial College London, Department of Materials; project: using protein language models to mimic natural ligands
- Personal compensation included

NSF I-Corps Regional Program, Northeast Hub; $750 May – Jun 2026
- Competitive cohort & customer-discovery project funding for Axioma Therapeutics

Grace Hopper AI Research Institute grant; $3,330 May – Aug 2025
- NJIT, computational protein design course development

Honors College Moonshot Grant for Engaged Scholarship; $6,500; 1 recipient per annum Jun – Jul 2025
- Male-contraceptive project
- Personal compensation included

Male Contraceptive Initiative Youth Advisory Board project grant; $1,000; 1 recipient in 2025 Jan – Jun 2025
- Grantee-led male-contraceptive project
- Personal compensation included

Male Contraceptive Initiative travel award; $500; only trainee selected May 2025
- National Contraception Meeting, Durham, North Carolina

NJIT Undergraduate Research & Innovation Student Seed Grant; $3,000 Oct 2024 – May 2025
- Phase-2 longitudinal research project, cyclic peptide therapeutic for multiple sclerosis

Honors Summer Research Institute (HSRI) grant; $3,000 Jun – Jul 2024
- ADHC & NJIT URI; multiple sclerosis & male-contraceptive projects
- Personal compensation included

First place, Honors Interdisciplinary Research Forum Fall 2025
- Male-contraceptive poster, NJIT Albert Dorman Honors College

Academic Excellence, Honors Merit, Honors Residential scholarships 2023 – 2027
- NJIT merit scholarships, part of award package exceeding $100,000

Highlander Achievement Scholarship – NJIT 2023 – 2027

Fellowships & selective programmes

Emerging Scholars Research Fellowship 2026
- Harvard Sikh Center; early modern Sikh history

Honors Summer Research Institute Fellowship – Fellow Jun 2024 – Jul 2024, Jun 2025 – Jul 2025

Goldwater Ambassadors Program Aug 2025 – Present
- Selected mentee, mentored by Goldwater Scholars; programme for underserved or underrepresented students

National Science Foundation Accelerating Research Translation Integrated Student Ambassador Network (ARTISAN); 1 of 15 fellows Apr 2025 – Present
- Selected/inducted on the basis of translational-research engagement

No Patient Left Behind Fellowship – Fellow Jun – Aug 2024

American College of Cardiology – Young Scholar May 2023 – Apr 2024

Skills & Languages

Computational biology	Rosetta; RosettaScripts; PyMOL; GROMACS & GROMACS scripting; NAMD; VMD; OpenMM; CHARMM; AlphaFold; Boltz; ESMFold; protein language models
Methods	Computational peptide design; docking; molecular dynamics; statistical mechanics; RMSD analysis; mathematical & stochastic modeling
Laboratory	Surface plasmon resonance (SPR); general dissection
Data & tools	Python; LaTeX; molecular dynamics trajectory analysis
Languages	English (C2); Punjabi (C1); Swedish (B2); Spanish (B1); Russian (B1)

Clinical & Translational

Emergency medical services

Emergency Medical Technician

Public Relations Chair, NJIT First Aid Squad Jan 2024 – Present
Senior Squad Member; former Cadet Crew Chief, Marlboro First Aid Squad Apr 2022 – Jan 2025

- Licensed New Jersey EMT; over 2,000 hours handling emergency medical situations across EMS roles.
- Represented the squad in presentations to NJIT senior leadership & Board of Trustees.

Public health education

Stop the Bleed Instructor

NJIT; ACS-certified instruction Oct 2025 – Present
New Jersey gurdwaras (hybrid English–Punjabi); NJ Office of Homeland Security & Preparedness; Sikh Youth Alliance Jan – May 2026

- Taught hemorrhage-control skills to 100+ people across NJIT & gurdwara outreach.
- Bilingual delivery because those most likely to be first on scene in a gurdwara are not always the most comfortable in English.

Commercialisation

Co-founder & Technical Lead May 2026 – Present
Axioma Therapeutics; funded through NSF I-Corps Northeast Hub

- Led customer discovery & lean-startup validation for peptide & mini-protein therapeutics targeting degenerative eye diseases.

Academic Service

Editorial & peer review

22nd Editor-in-Chief; former Associate Editor Dec 2024 – Present
Journal of Young Investigators (JYI)

- Led oldest international undergraduate research journal, funded & featured by Nature, Science, NSF & the Burroughs Wellcome Fund; earlier associate-editor work spanned biology, physical sciences & social sciences.
- Developed structured, multi-stage hiring, asynchronous onboarding, tiered training, & a shared manuscript workflow board to reduce institutional knowledge loss; re-established translations, bolstered peer review, restored perspective articles, & developed partnerships for undergraduate research events.
- 2025–26: 27 new staff from 96 applications; two associate editors & a research editor assigned to each manuscript, with a two-week review window; author/reviewer AI policy published.

Teaching & mentorship

Research Lead Sep 2026 – May 2027
Governor's STEM Scholars; Research & Development Council of New Jersey (Expected)

- Mentor high-school team on a year-long engineering & research project aligned to a UN Sustainable Development Goal & a New Jersey community need; oversee meetings, budget, milestones & deliverables, with final presentation planned across four statewide conferences; $1,000 stipend.

Ethics Coordinator, Mentor (Neuroscience & Biology), Admissions Officer, Symposium Organizer & Grant Writer; former Ethics Curriculum Developer May 2024 – Present
Science Mentorship Institute (Sci-Mi)

- Contributed to ethics instruction & research mentoring; delivered 2026 lecture series on ethics in science & research.

Teaching Assistant, Honors First Year Seminar, Albert Dorman Honors College, NJIT — Sep 2024 – Dec 2024, Sep 2025 – Dec 2025, Sep 2026 – Dec 2026

Teaching Assistant, Computational Protein Design, Grace Hopper AI Research Institute, NJIT — May – Aug 2025

Sikh History (Virsa) Teacher, Khalsa School, Sikh Sabha of New Jersey — Sep 2020 – Present

Research-team mentor & computational instructor, KumarLab; 15+ trainees — Oct 2023 – Present

National & professional

Research Chair Jun 2025 – Present
American Association of Physicians of Indian Origin (AAPI) PreMed Board

- Lead research-focused committee & programming to support students, researchers & physicians.
- Helped organize & host 2025 & 2026 Fall Virtual Research Symposia; sole MC in 2025 & co-MC in 2026, with 11 US & international presentations & seven physician judges recruited.

Board Member & Grant Reviewer, Male Contraceptive Initiative (MCI) Youth Advisory Board — Aug 2025 – Present

Member, Pre-Medical Committee, American Physician Scientists Association (APSA) — Mar 2026 – Present

University

Senior Representative, Honors Scholar Council — Sep 2026 – Present

Member, Student Conduct Board Dec 2024 – Present
NJIT

- Adjudicate academic & behavioral conduct cases alongside faculty, staff & university administrators.

Chair; former Chair-in-Training, Highlander Integrity Council, NJIT — Sep 2024 – Present

Member, Inclusive Excellence Student Advisory Board, NJIT — Feb 2025 – Present

Member, Multifaith Council, NJIT — Jan 2025 – Present

Organizer, 'It's Different for Girls' screening & panel, NJIT ADHC & CSLA co-sponsored colloquium — Mar 2026

University Representative, BioNJ 33rd Annual Dinner Meeting & Innovation Celebration — Feb 2026

Steering Committee Lead, Albert Dorman Honors College First Year Retreat — Mar – Aug 2025

Advisor; former President & Secretary, Honors Hall Council, NJIT — Feb 2024 – May 2025

Honors College Representative, National Collegiate Honors Council Annual Conference — Nov 2024

Leadership & Community

Vice President; former President & Events Coordinator Dec 2023 – Present
Sikh Association of NJIT (SANJIT)

- Vice President, 2026 – present; President, 2023 – 2026; earlier Events Coordinator.

Co-founder; former President & Treasurer, NJIT MedicZero — Nov 2023 – May 2026

Co-founder, Delta Epsilon Mu, pre-medical professional fraternity — Feb 2024 – May 2025

Youth Camp Organizer & Coordinator, Sikh Sabha of New Jersey — Aug 2021 – Present

Youth Camp Counselor, Guru Nanak Sikh Heritage — Aug 2021 – Aug 2025

Content Manager, The Mardana Project — May 2026 – Present

Certifications

- New Jersey Emergency Medical Technician licence
- American College of Surgeons Stop the Bleed certification & instructor licence
- NIH, Introduction to the Principles & Practice of Clinical Research (IPPCR), certificate of completion
- OSHA General Industry Safety & Health certification
- CITI Biomedical Research certification
- CITI Responsible Conduct of Research (RCR) Basic certification
- American Heart Association BLS & CPR, Adult/Child/Infant

Supplemental Interests

Gurmat Sangeet (classical Indian music in the Sikh devotional tradition); wrestling & martial arts; ping-pong; travel; reading history, epistemology & comparative theology.

References available upon request